Laws of Motion

1) A force of 250 N is applied on a table from one side and a force of 300 N is applied on the same table in the opposite direction along the same straight line. How much is the resultant force on the table?
(550 N, 300 N, 50 N, 250 N)
Answer:
50 N

2) A trolley is pulled towards east with a force of 500 N and towards west with a force of 750 N. What is the resultant force felt on the trolley?
(1250 N to east, 1250 N to west, 250 N to east, 250 N to west)
Answer:
250 N to the west

3) A cart is pushed by team A and team B towards East with forces 1500 N and 2000 N respectively. What is the resultant force felt on the car?
(1500 N, 2000 N, 500 N, 3500 N)
Answer:
3500 N

4) A table is pushed towards the east with a force of 1000 N and is pulled in the opposite direction with a force of 300 N. How much is the resultant force felt on the table?
(700 N to east, 1300 N to east, 700 N to west, 1300 N to west)
Answer:
1300 N to the south

5) In which of the following situations is the resultant force zero?
a) A force of 100 N is applied to the East and a force of 200 N to the West.
b) Pushing a body with a force of 100 N and pulling it with a force of 100 N from the opposite direction.
c) A horse pulls the cart forward with a force of 400 N and the horse cart pulls the horse backward with a force of 400 N.
Answer:
c

6) Which force can change the state of an object at rest?
(balanced force, unbalanced force, resultant force)
Answer:
unbalanced force

7) Unbalanced forces cannot change ..... state of an object.
(motion, rest, velocity, solid state)
Answer:
solid state

8) Internal forces are always .....
(balanced force, unbalanced force, resultant force)
Answer:
balanced force

9) What will not happen if a force of 100 N is applied to a moving object?
(Speed of motion increases, speed of motion decreases, direction of motion changes, weight increases)
Answer:
weight increases

10) A person pushes a car by sitting inside a car.
a) The car does not move. Why?
b) Justify your answer.
Answer:
a) The force applied to the car is balanced.
b) All internal forces are balanced forces.

11) In Galileo's marble and channel experiment,
a) Write down the reason why the marble tried to reach the maximum height.
b) Why does the marble come to rest after rising to some height?
Answer:
a) Due to inertia
b) Due to the force of friction

12) a) Which motion does not require an unbalanced external force?
(Uniform rectilinear motion, To change direction of a body moving at a constant speed, To bring an object to rest)
b) Which scientist's experiments helped you reach the above conclusions?
Answer:
a) The body travelling along a straight line with uniform speed.
b) Galileo Galilei

13) State Newton's first law of motion.
Answer:
Newton's first law of motion.

14) a) What is the unit of force?
b) Write down a definition of force.
Answer:
a) newton (N)
b) Force – Definition

15) A force of 400 N is applied to a table to the east and 600 N to the west along the same straight line.
a) What is the effective force felt on the table?
b) How are effective and resultant forces related?
c) In which direction will the table move? How much force is used to move the table?
Answer:
a) 200 N to the west
b) Both are the same
c) to the west. 200 N.

16) Three people are standing in a jeep and are pushing the jeep.
a) Will the jeep move?
b) What is the resultant force on the jeep?
c) Justify your answers to questions a and b.
Answer:
a) No
b) Zero
c) a- All internal forces are balanced forces. b- Balanced force cannot produce motion.

17) a) If the resultant force felt on an object is zero then by what name are the applied forces known?
b) Can such forces change the state of motion of the object? Why?
Answer:
a) Balanced force
b) No. Balanced force cannot produce motion. Unbalanced force alone can produce motion.

18) a) An object is not capable of changing its state of rest or motion by itself. By what name is this property known?
b) What must be applied to the object to change this particular state of the object?
c) State the law relating to questions a and b. Who formulated this law?
Answer:
a) Inertia
b) Unbalanced external force. Newton.

19) In Galileo's marble and channel experiment,
a) Write down the reason why the ball tries to reach its maximum height.
b) Why does the ball come to rest after rising a little?
c) What knowledge did you gain from Galileo's marble and channel experiment?
Answer:
a) Inertia
b) Due to inertia
c) An unbalanced external force is to be applied in the opposite direction of motion of a body to bring it to rest. An unbalanced external force is not required to maintain uniform motion along a straight line.